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Dynamical calibration of a metallicity-dependent stellar mass–luminosity relation from Gaia wide binaries

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ABSTRACT

Context. Wide binaries provide a dynamical mass scale that does not depend on stellar models, but the mass–luminosity relation (MLR) it constrains depends on metallicity, which is poorly known for faint dwarfs.
Aims. We calibrate a metallicity-dependent MLR of main-sequence stars from the relative orbital motion of Gaia wide binaries.
Methods. Metallicities of both components come from J-CaPS XP spectra and a shared-metallicity prior for each pair. The MLR is a monotone surface in absolute G magnitude and [M/H] relative to PARSEC, fitted with a selection-corrected dynamical likelihood.
Results.

Key words. binaries: visual – stars: low-mass – stars: fundamental parameters – methods: statistical – Galaxy: solar neighbourhood

1. Introduction

In this paper, we calibrate the mass–luminosity relation (MLR) of main-sequence stars over the absolute magnitude range $3.5 \leq M_G \leq 13.5$ as a function of metallicity. At solar metallicity this range corresponds to masses of about 0.1 to $1.3\,M_\odot$, from late-F and G dwarfs down to the bottom of the main sequence, and our focus is on the K and M dwarfs that make up most of it.

Stars of low mass are the numerically dominant stellar population. M dwarfs make up roughly three quarters of the stars within 10 pc of the Sun (Henry et al. 2018), and an initial mass function that is flat or declining below $\sim 0.5\,M_\odot$ (Kroupa 2001; Chabrier 2003) still places a large part of the stellar mass of the Galactic disk in stars below one solar mass (Chabrier 2001). They therefore hold much of the baryonic mass that has been locked up in stars, and the mass function inferred from star counts, and every quantity derived from it (the mass-to-light ratio of stellar populations, the contribution of dwarfs to the dynamical mass budget of the disk, the claimed variations of the IMF with environment and metallicity), depends directly on how masses are assigned to dwarfs. The assignment is not innocuous: a mass–luminosity relation that is wrong in one photometric band propagates into the local mass function derived from the luminosity function in that band (Delfosse et al. 2000). Low-mass stars are also the most common hosts of small planets (Dressing & Charbonneau 2015). The radius of a transiting planet scales with the stellar radius, which for a dwarf is obtained from its mass or luminosity (Mann et al. 2015), the semi-amplitude of its radial-velocity signal scales as $M_\star^{-2/3}$, and the distance of the habitable zone depends on the stellar luminosity and thus again on mass. Errors of a few per cent in the stellar mass pass into the planetary mass, radius and bulk density.

Stellar mass is almost never a measured quantity. Asteroseismology provides an independent route for single stars: the frequency of maximum power and the large frequency separation of solar-like oscillations give the mass and radius of a star without a companion, and have been applied to thousands of red giants and to hundreds of dwarfs and subgiants (Chaplin & Miglio 2013; Serenelli et al. 2017; Pinsonneault et al. 2018). The method is limited for our purpose. The scaling relations rely on the effective temperature and on model-based corrections, and their calibration to dynamical masses in clusters leaves systematic differences that depend on evolutionary state and metallicity (Pinsonneault et al. 2018). More importantly, the oscillation amplitude decreases steeply towards lower luminosity (Kjeldsen & Bedding 1995), so the detections are concentrated in giants and in dwarfs hotter than about 5000 K, and K and M dwarfs are essentially out of reach (Chaplin & Miglio 2013). The giants, which are the bulk of the asteroseismic sample, are evolved and cannot calibrate a main-sequence MLR.

Binaries are therefore the main place where stellar masses are calibrated, and the methods differ in what they measure. In eclipsing binaries with double-lined spectroscopic orbits, the radial-velocity amplitudes and the orbital inclination from the eclipse light curve give the mass of each component directly, to a few per cent or better (Torres et al. 2010). The systems are, however, close, with periods of days, so tidal interaction and fast rotation may alter the stars, and low-mass eclipsing binaries are among those that show the radius and temperature anomalies discussed be-

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low. Visual and astrometric orbits of nearby binaries determine the total mass from Kepler's third law once the period, the semi-major axis and the parallax are known, and the mass ratio from the motion of the components about the barycentre. Combined with long-term imaging and adaptive optics this is the basis of the best relations for M dwarfs (Henry & McCarthy 1993; Delfosse et al. 2000; Benedict et al. 2016; Mann et al. 2019). It needs monitoring over years to decades, requires systems that are close enough to be resolved, and has produced samples of a few tens of systems. Spectroscopic binaries without eclipses give the projected orbit only. For double-lined systems (SB2) the radial velocities of both stars yield $M \sin^3 i$ for each component, and for single-lined systems (SB1) only the mass function, a combination of the two masses and the inclination that cannot be inverted without an assumption on the mass of the visible star. In both cases the inclination is unknown, unless it is supplied by an astrometric orbit. The Gaia astrometric and spectroscopic orbits (Gaia Collaboration et al. 2023) make such combinations possible for large numbers of binaries and have been used to derive the MLR in the G band down to $0.12\,M_\odot$ (Chevalier et al. 2023), but the solutions are restricted to periods of the order of the mission length and to systems with a resolvable photocentre motion, and the SB1 solutions depend on an adopted mass of the primary. All these methods apply to close or moderately wide systems with orbital periods of days to decades, and none of them reaches large samples across the colour–magnitude diagram.

For all other stars the mass has to be inferred from an observable that correlates with it, usually the absolute magnitude, through a calibrated MLR, and because the calibrating systems are rare the empirical relations rest on a few tens of systems. The near-infrared relation of Mann et al. (2019), the most precise, is based on 62 binaries and yields masses to 2–3%. It finds no measurable dependence on metallicity at the solar-neighbourhood level, but this holds for the K_s band and for a narrow metallicity range, and the dependence in bands that are more sensitive to molecular absorption, such as the optical bands of Gaia, is not constrained in the same way.

For the large photometric samples of Gaia the relation is usually taken from stellar evolution models, whose predictions for low-mass stars remain uncertain. The interior structure depends on the treatment of convection, for which mixing-length theory has to be calibrated or replaced by fits to multidimensional simulations, and on the equation of state and opacities at high density (Chabrier & Baraffe 2000; Baraffe et al. 2015). The emergent flux, and hence the colours and bolometric corrections that connect a model luminosity to a Gaia magnitude, are governed by molecular opacities (TiO, VO, H_2O) whose line lists are incomplete, so that models have long predicted optical colours that disagree with observed M dwarfs (Baraffe et al. 2015). The PARSEC models required an empirical shift of the temperature–optical-depth relation of the atmosphere, calibrated on the observed mass–radius relation, to reproduce the lower main sequence (Chen et al. 2014). Eclipsing binaries show in addition that low-mass stars have radii larger and effective temperatures lower than predicted by non-magnetic models at the same mass, a discrepancy that is attributed to magnetic activity and spots but is not yet fully explained (Torres & Ribas 2002; Feiden & Chaboyer 2012; Spada et al. 2013; Feiden & Chaboyer 2014). These effects change both the luminosity and the colour at fixed mass, and the effect on the MLR depends on band and on metallicity, because metallicity controls the opacity of the atmosphere and the interior. The discrepancy between models and observations is largest in the visual bands and decreases in the near-infrared (Delfosse et al. 2000). A calibration in the Gaia bands as a function of metallicity, which can be compared with model predictions, is therefore missing.

Wide binaries offer a way to obtain such a calibration for large samples. Their projected separations are 10^3–10^4 au and their orbital periods 10^4–10^6 yr, so no orbit can be followed, and each pair provides a single snapshot: the projected separation s and the plane-of-sky velocity difference Δv of the two components. Kepler's law nevertheless relates these quantities to the total mass, $\Delta v\sqrt{s} \propto \sqrt{M_{\rm tot}}$, with a proportionality that depends on the orbital phase, eccentricity and inclination of the pair. For a population with random phases, orientations and a known eccentricity distribution, the distribution of $u = \Delta v\sqrt{s}$ has a fixed shape whose scale is set by the mass, so the mass of a group of pairs can be recovered statistically. Gaia has made this practical: its astrometry gives the proper-motion difference and the parallax of hundreds of thousands of pairs (El-Badry et al. 2021). The same relative velocities have been used to test gravity at the low accelerations of wide pairs, where the stellar masses are taken from a relation and the shape of the distribution of Δv is compared with Newtonian and modified-gravity predictions (Hernández et al. 2012; Chae 2023; Banik et al. 2024); the result depends on how the hidden third stars are treated and remains debated. The inverse problem, in which Newtonian dynamics is assumed and the mass is inferred, has been applied to calibrate an MLR in the $G_{\rm RP}$ band with a few thousand pairs (Giovinazzi & Blake 2022).

Hwang et al. (2024) developed the most general version of this idea. They model the mass of a star as a function of its colour and absolute magnitude with a neural network, trained by maximising the likelihood of u for the wide binaries of the El-Badry et al. (2021) catalogue, with a mixture component for high-u outliers. The masses depend only on Newtonian dynamics, and they cover the whole Hertzsprung–Russell diagram: main-sequence stars from 0.1 to $2\,M_\odot$, the sequences of unresolved binaries and triples above the main sequence, the pre-main-sequence stars, and the mean masses of giants and white dwarfs. This work showed that Gaia wide binaries are a model-independent mass scale and that dynamical masses agree with isochrones better than expected from the systematics of the models, with a preference for PARSEC over MIST at low masses. The purpose of that study was a global map of the diagram, and its output is the mean mass of the stars at each position in the colour–magnitude diagram, whichever population occupies it: the single stars, the unresolved binaries and triples, and the pre-main-sequence stars overlap on the lower main sequence, and the single-star relation is obtained along a running median of the sequence. The input has no metallicity, which the authors identify as a natural extension, and the uncertainty is the scatter of repeated network predictions with dropout, not a posterior that includes the measurement errors, the outlier model and the selection of the sample. The cut on the significance of the proper-motion difference is satisfied through the choice of distance and separation limits but not modelled in the likelihood.

Here we build on this approach and turn it into a calibration of the MLR of single main-sequence stars. We use the pairs of the El-Badry et al. (2021) catalogue for which J-CaPS metallicities (Li et al. 2026) are available for both components. Gaia XP spectra allow us to identify and remove unresolved binaries, so that the sample consists almost entirely of true wide pairs of single main-sequence stars and the relation refers to a well-defined population rather than to the mean of all stars at a position in the diagram. Metallicity is a variable of the model: we assume that the components of a wide binary share the same metallicity, combine the two measurements into a posterior for each system, and fit a monotone MLR surface in absolute G magnitude and metallicity, defined relative to PARSEC (Bressan et al. 2012). The likelihood for u includes the measurement errors, an outlier component, and the truncation produced by the catalogue's cut on the significance of the proper-motion difference. We sample the posterior of the relation with Hamiltonian Monte Carlo (NUTS), which gives credible intervals that include the uncertainty of the data, the metallicity and the nuisance parameters. (Result summary: offset from PARSEC as a function of M_G, comparison with Mann et al. 2019, metallicity dependence, comparison with Hwang et al. 2024; to be written when the results are final.)

The paper is organised as follows. Section 2 describes the data, Sect. 3 the inference, Sect. 4 the resulting MLR, and Sect. 5 the tests of the main modelling assumptions. We discuss the results and their limitations in Sect. 6 and conclude in Sect. 7.

2. Data

2.1. Wide-binary catalogue and sample selection

We apply quality cuts to ensure reliable astrometry and photometry. The selection criteria are summarised in Table 1.

We calculate u for each wide binary system of the El-Badry et al. (2021) catalogue from Gaia DR3 observables. We require `R_chance_align` < 0.1 and $s < 10^4$ AU to eliminate most chance alignments, $\varpi > 1$ mas, `Fidelity_v2` ≥ 0.5 to ensure stars with reliable astrometry, and the single-star criterion based on Gaia XP spectra from Li et al. (202?), $\Delta\mathrm{BIC} > 0$. We further require Gaia flux signal-to-noise ratios > 10, $|C^*| \leq 3\sigma_{C^*}$ to remove sources with poor BP/RP observations, a blended-transit fraction $\beta > 0.5$ to remove sources affected by blending, and a relative-velocity signal-to-noise ratio $u/\sigma_u > 3$.

2.2. J-CaPS metallicities

2.3. Observed dynamical statistic

3. Method

3.1. Overview

The inference is modular. Stage 1a uses the metallicity differences between the two components of each binary to estimate the relative calibration of the J-CaPS metallicities. Conditional on this estimate, Stage 1b combines the J-CaPS measurements of both components with the colour–magnitude diagram (CMD) of stars with reliable model colours and a hierarchical population distribution, and returns a posterior probability distribution for the shared metallicity Z_j of every binary. Stage 2 takes these distributions as fixed inputs, predicts the masses of both components from a metallicity-dependent MLR at every metallicity, and evaluates the wide-binary dynamical likelihood, corrected for the selection of the sample by the significance of the proper-motion difference. The dynamical data do not feed back into Stage 1. The workflow is shown in Fig. 1.

Binary systems are indexed by j, their components by $k \in \{1, 2\}$ and metallicity nodes by q. Hats denote observed quantities. x_{jk} is the Gaia absolute magnitude M_G of a component, C_{jk} its dereddened colour $(G_{\mathrm{BP}} - G_{\mathrm{RP}})_0$, $\widehat{z}_{jk}$ the J-CaPS metallicity and $\sigma_{z,jk}$ its reported error. The metallicity is expressed on the PARSEC scale, $Z \equiv [\mathrm{M/H}]$. The J-CaPS inputs are the raw measurements and the reported errors. Catalogue columns that already apply the equal-composition assumption of the binary are not used, because they would count that information twice in the likelihood.

3.2. System metallicities

Relative calibration (Stage 1a). Each star carries an independent Student-t measurement of the shared metallicity,

$$\widehat{z}_{jk} \mid Z_j \sim \mathcal{T}_\nu\Big[Z_j + \delta_z + b_G(x_{jk}),\ \sqrt{\sigma_{z,jk}^2 + s_z^2}\Big] \equiv \mathcal{L}_{jk}^{\mathrm{spec}}(Z_j), \quad (1)$$

where $b_G(x)$ is a magnitude-dependent relative bias, s_z an additional scatter common to all stars, ν the degrees of freedom, and δ_z a zero-point offset between J-CaPS and PARSEC. Conditional on Z_j the two components are independent. The relative parameters (b_G, s_z, ν) are determined from the differences between the two stars of each binary, in which Z_j and δ_z cancel, by maximum likelihood, and are held fixed in Stage 1b.

Absolute calibration and system posteriors (Stage 1b). The absolute metallicity scale is set by the CMD. For stars warm enough for the model colours to be reliable, PARSEC (Bressan et al. 2012) predicts the colour $C_{\mathrm{P}}(x, Z)$ and the effective temperature $T_{\mathrm{eff,P}}(x, Z)$ of the main sequence. We average the isochrones with $9.3 < \log_{10}(\mathrm{age/yr}) < 10$ at fixed M_G and metallicity, so age is not a latent variable, and tabulate the relation on a grid in $3.5 \leq M_G \leq 13.5$. With the formal colour uncertainty $\sigma_{C,jk}$ from the BP and RP signal-to-noise ratios, the colour likelihood is

$$C_{jk} \mid Z_j \sim \mathcal{T}_4\Big[C_{\mathrm{P}}(x_{jk}, Z_j),\ \sqrt{\sigma_{C,jk}^2 + s_C^2}\Big] \equiv \mathcal{L}_{jk}^{\mathrm{CMD}}(Z_j), \quad (2)$$

with an excess scatter s_C and a colour zero point fixed to zero. A component enters this term only if it is warm over the whole metallicity domain, $m_{jk} = \mathbb{I}[\min_q T_{\mathrm{eff,P}}(x_{jk}, Z_q) \geq 4000\,\mathrm{K}]$, so that the membership does not depend on the proposed metallicity. Components with $m_{jk} = 0$ contribute only their J-CaPS likelihood.

The metallicity is evaluated on a grid of 81 nodes Z_q between -1.0 and 0.6 dex. The population distribution is a mixture of six truncated Gaussians with equally spaced

Table 1. Quality cuts applied to the wide binaries of the El-Badry et al. (2021) catalogue.

Criterion	Cut	Purpose
`R_chance_align`	< 0.1	Eliminate chance alignments
s (projected separation)	$< 10^4$ AU	Eliminate chance alignments
ϖ (parallax)	> 1 mas	Targeted distance volume
`Fidelity_v2`	≥ 0.5	Stars with reliable astrometry
ΔBIC	> 0	XP-based single-star classification (Li et al. 202?)
Gaia flux S/N	> 10	High-quality photometry
$\lvert C^* \rvert$	$\leq 3\sigma_{C^*}$	Good BP/RP observations
β (blended-transit fraction)	> 0.5	Remove blended sources
u/σ_u (relative-velocity S/N)	> 3	Reliable velocity measurements

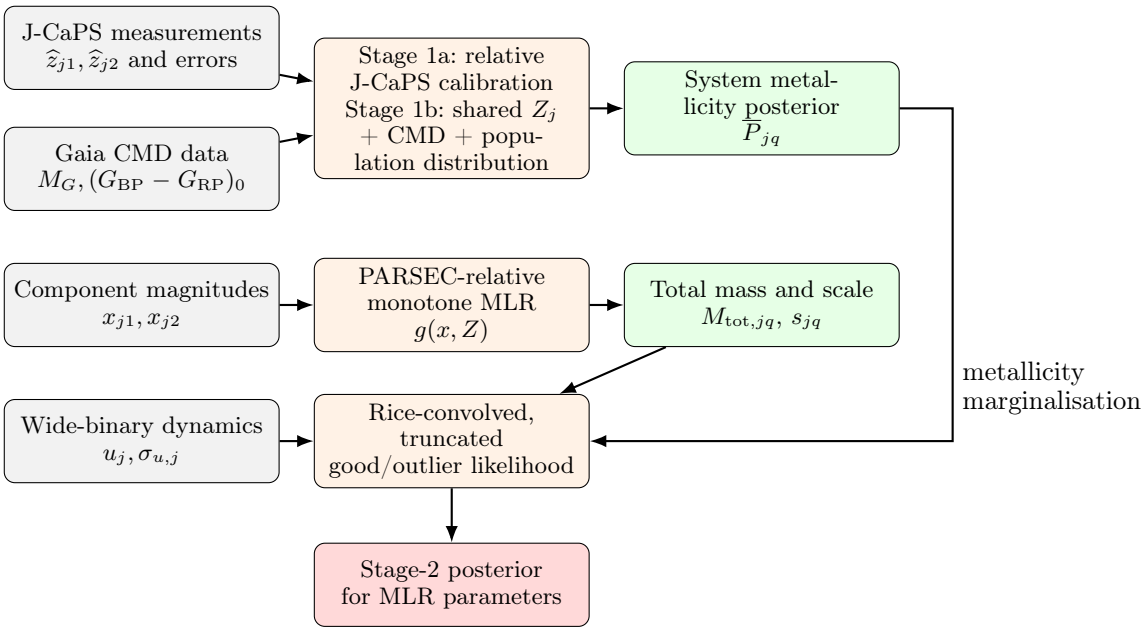


Fig. 1. Inference workflow. Stage 1a estimates the relative J-CaPS calibration from within-binary differences. Conditional on these estimates, Stage 1b returns a posterior probability mass over the shared metallicity of each system. Stage 2 propagates this uncertainty into the dynamical MLR fit but does not update either Stage 1 module.

centres on this interval, a common width equal to the interval divided by six, and weights $\boldsymbol{\omega}$ with a flat Dirichlet prior. With $\boldsymbol{\psi} = (\delta_z, s_C, \boldsymbol{\omega})$ the global parameters, the unnormalised density of one system is

$$\widetilde{p}_j(Z, \boldsymbol{\psi}) = p_{\rm pop}(Z \mid \boldsymbol{\omega})\, p(\boldsymbol{\psi}) \times \prod_{k=1}^{2} \mathcal{L}_{jk}^{\rm spec}(Z) \left[\mathcal{L}_{jk}^{\rm CMD}(Z)\right]^{m_{jk}}, \quad (3)$$

and the global parameters are inferred from the product of the marginal evidences of all systems. The posterior probability mass of system j at node Z_q, with the global parameters fixed at their posterior mean, is $\overline{P}_{jq}$. This single distribution per system is the only metallicity information passed to Stage 2.

3.3. A PARSEC-relative monotone MLR

We describe the mass of a star of absolute magnitude x and metallicity Z by the surface $g(x, Z) = \log_{10}[M/M_\odot]$. The surface is a tensor-product cubic B-spline (de Boor 2001) with 8 coefficients in x and 4 in Z,

$$g(x, Z) = \sum_{i=1}^{8} \sum_{h=1}^{4} \Theta_{ih} B_i(x) C_h(Z), \quad (4)$$

with clamped knots at $x = (3.5, 5.5, 7.5, 9.5, 11.5, 13.5)$ mag and $Z = (-1.0, 0.6)$ dex. The coefficients are written $\Theta = \Pi + D$, where Π is the projection of the PARSEC mass surface $g_{\rm P}(x, Z)$ onto the same basis and D is the correction fitted to the data. The mass relation is then reported relative to PARSEC through $\Delta_{\rm P} = g - g_{\rm P}$, which includes the small projection error (RMS 0.002 dex). We impose two physical constraints: a fainter star at fixed metallicity is not more massive, and a more metal-rich star at fixed magnitude is not less massive,

$$\Theta_{i+1,h} \leq \Theta_{ih}, \qquad \Theta_{i,h+1} \geq \Theta_{ih}. \quad (5)$$

These inequalities on the B-spline coefficients imply $\partial g/\partial x \leq 0$ and $\partial g/\partial Z \geq 0$ everywhere in the domain. They apply to the total surface and not to D, so the correction itself may have either sign and the data are free to move the relation away from PARSEC. No constraint is placed on

the mixed differences. The inequalities are enforced by construction through a reparametrisation with unconstrained variables (Appendix A).

3.4. Dynamical likelihood

For a bound pair of total mass $m_{\rm tot}$, the instantaneous relative speed scales as $\sqrt{Gm_{\rm tot}/a}$ with a factor that depends on eccentricity and orbital phase. We use the statistic

$$u = v\sqrt{s}, \tag{6}$$

where v is the projected relative speed and s the projected separation. Because $v \propto a^{-1/2}$ and $s \propto a$, the semi-major axis cancels, and $u \propto \sqrt{m_{\rm tot}/M_\odot}$ with a factor that depends only on the eccentricity, the phase and the orientation. The mass-scaled statistic $\widetilde{u} = u/\sqrt{m_{\rm tot}/M_\odot}$ therefore has a distribution $p(\widetilde{u})$ that does not depend on mass, and a noiseless measurement has the density $p(u \mid m_{\rm tot}) = p(u/s)/s$ with $s = \sqrt{m_{\rm tot}/M_\odot}$. We write $s_{jq} = \sqrt{M_{{\rm tot},jq}/M_\odot}$ for pair j at metallicity node q, with $M_{{\rm tot},jq} = M_\odot[10^{g(x_{j1},Z_q)} + 10^{g(x_{j2},Z_q)}]$.

The observed u_j is the magnitude of a two-dimensional velocity difference with Gaussian errors of common scale $\sigma_{u,j}$, so for a latent speed v it follows the Rice distribution (Rice 1945),

$$R(u_j \mid v, \sigma_{u,j}) = \frac{u_j}{\sigma_{u,j}^2} \exp\left[-\frac{u_j^2 + v^2}{2\sigma_{u,j}^2}\right] I_0\left(\frac{u_j v}{\sigma_{u,j}^2}\right), \tag{7}$$

with I_0 the modified Bessel function of order zero and $\sigma_{u,j} = u_j/(\Delta\mu/\sigma_{\Delta\mu})_j$ from the proper-motion difference and its uncertainty. Pairs that are not described by a bound binary of two stars, for example because of an unresolved third body, are included through an outlier component.

The good component uses the functional form of Hwang et al. (2024),

$$p(\widetilde{u}) = \frac{1}{N}\,\widetilde{u} \exp\left[-B\widetilde{u}^2 - \exp\left(\frac{\widetilde{u} - u_c}{C}\right)\right] \times \mathbb{I}(0 < \widetilde{u} \le 80), \tag{8}$$

with $B = 2.5050 \times 10^{-3}$, $u_c = 37.2461$ and $C = 2.4500$, and N normalising p on $(0, 80]$. The constants are fitted to a Monte Carlo of Keplerian orbits with uniform mean anomalies, isotropic orientations and eccentricities drawn from $p(e) \propto e^{\alpha_e}$, in which each simulated pair is weighted so that its joint distribution of projected separation and distance matches the sample (Appendix C). We adopt $\alpha_e = 1.23$, the separation-dependent power-law index of Hwang et al. (2022) evaluated at the projected separations of our pairs. That index is measured from the angle between the separation and relative-velocity vectors of Gaia wide binaries and does not involve stellar masses. The shape is held fixed, and the effect of α_e on the MLR is quantified in Sect. 5.1. Its upper edge is set by energy conservation, $\widetilde{u} < \sqrt{2GM_\odot/{\rm AU}} = 42.12\,{\rm km\,s^{-1}}\sqrt{\rm AU}$ for any eccentricity distribution. The outlier component is a Gaussian of mean 40 and standard deviation $13\,{\rm km\,s^{-1}}\sqrt{\rm AU}$ in the latent speed, truncated and normalised on $[0, 80]\,{\rm km\,s^{-1}}\sqrt{\rm AU}$,

$$q(v) = \frac{\phi[(v-40)/13]}{13\{\Phi[(80-40)/13] - \Phi(-40/13)\}} \times \mathbb{I}(0 \le v \le 80), \tag{9}$$

where ϕ and Φ are the standard normal density and distribution function. It does not depend on the mass of the pair. The likelihood of pair j before selection is

$$\mathcal{L}_j(s) = (1 - f_{\rm outlier}) \int_0^{80} p(\widetilde{u})\, R(u_j \mid s\widetilde{u}, \sigma_{u,j})\, {\rm d}\widetilde{u} + f_{\rm outlier} \int_0^{80} q(v)\, R(u_j \mid v, \sigma_{u,j})\, {\rm d}v, \tag{10}$$

where the first integral carries the Jacobian $1/s$ of the change of variable from v to $\widetilde{u}$ and $f_{\rm outlier}$ is a free parameter.

3.5. Selection and marginalisation over metallicity

The catalogue retains only pairs whose proper-motion difference is detected at more than $c = 3$ times its uncertainty. Since $\sigma_{u,j}$ is fixed for each pair by the astrometric uncertainties, the parallax and the projected separation, this is a threshold on the observed statistic,

$$u_j > c\,\sigma_{u,j}. \tag{11}$$

Pairs with a small latent speed and a large $\sigma_{u,j}$ are preferentially removed, so the observed u_j are not draws from Eq. (10). The density of u_j conditional on passing the cut is

$$\mathcal{L}_j^{\rm sel}(s) = \frac{\mathcal{L}_j(s)}{\mathcal{A}_j(s)}, \tag{12}$$

where $\mathcal{A}_j(s)$ is the probability that a pair of mass scale s satisfies Eq. (11). For a latent speed v the Rice distribution gives this probability as the Marcum Q function, $\Pr(u > c\,\sigma_{u,j} \mid v) = Q_1(v/\sigma_{u,j}, c)$, and averaging over the two components,

$$\mathcal{A}_j(s) = (1 - f_{\rm outlier}) \int_0^{80} p(\widetilde{u})\, Q_1\left(\frac{s\widetilde{u}}{\sigma_{u,j}}, c\right) {\rm d}\widetilde{u} + f_{\rm outlier} \int_0^{80} q(v)\, Q_1\left(\frac{v}{\sigma_{u,j}}, c\right) {\rm d}v. \tag{13}$$

The pass probability of the good component depends on the mass and the uncertainty only through $s/\sigma_{u,j}$ and increases with s; that of the outlier component does not depend on the mass. Dividing by $\mathcal{A}_j$ removes the probability that the untruncated model places below the cut, where no pair is observed. This probability is larger at low mass, so the correction lowers the inferred masses relative to Eq. (10). The fraction of outliers among the selected pairs differs from $f_{\rm outlier}$, which is the mixing fraction before selection.

The selection acts on the true mass of the pair, so $\mathcal{A}_j$ is evaluated at each metallicity node before the sum over the Stage-1b weights,

$$\mathcal{L}_j^{\rm sel}(\Theta) = \sum_q \overline{P}_{jq}\, \frac{\mathcal{L}_j(s_{jq})}{\mathcal{A}_j(s_{jq})}, \tag{14}$$

where s_{jq} follows from the MLR coefficients Θ through Eq. (4). The dynamical term of the Stage-2 log-posterior is $\sum_j \ln \mathcal{L}_j^{\rm sel}$.

3.6. Priors and posterior computation

The free parameters of Stage 2 are the 35 unconstrained coordinates of the monotone surface, the two increment scales λ_x, λ_Z of the reparametrisation, and $f_{\rm outlier}$; their priors are given in Appendix A. The prior on the first coefficient and on the increment scales is centred on the values of PARSEC, so the data are required to move the relation away from the model. Stage 1b and Stage 2 are sampled with the No-U-Turn Sampler (Hoffman & Gelman 2014), using four chains of 1 000 retained draws each. The component likelihoods of Eq. (10) are computed for every pair on a geometric grid of 1 024 values of s between 0.25 and 2.5, by Gauss–Legendre quadrature over the latent speed within $\pm 14\,\sigma_{u,j}$ of u_j, and interpolated in the logarithm during sampling. The pass probability $\mathcal{A}_j^{\rm good}$ is tabulated as a function of $s/\sigma_{u,j}$ with $Q_1(a,c) = \Pr[\chi^2_2(a^2) > c^2]$, the survival function of the non-central χ^2 distribution with two degrees of freedom.

4. Results

4.1. System metallicities

4.2. Mass–luminosity relation at solar metallicity

4.3. Metallicity dependence

4.4. Outlier component and goodness of fit

4.5. Effect of the selection correction

5. Systematic checks

5.1. Shape of $p(\widetilde{u})$

The shape of $p(\widetilde{u})$ is fixed by the orbital elements of the population, and of these only the eccentricity distribution is not determined by the sample itself: the phases and orientations are random, and the dependence on the separation and distance distributions is removed by the weighting of Appendix C. A more eccentric population spends a larger fraction of each orbit near apocentre, at large separation and low speed, so $p(\widetilde{u})$ moves to lower values and the masses inferred at fixed u increase. We repeat the full inference, including the per-pair likelihood tables, with the shapes for $\alpha_e = 1.0$ and 1.5 (Table C.1). The lower value is the thermal distribution. It allows for the bias of the measured index towards higher values that Hwang et al. (2022) attribute to their cut on the significance of the proper-motion difference, which at separations above $10^{3.5}$ au they estimate at about 0.1, and for the extrapolation from their sample within 200 pc to ours within 1 kpc. The upper value allows for unresolved companions, which randomise the measured velocity directions and bias the index towards the thermal value.

At [M/H] = 0, $\alpha_e = 1.0$ lowers the masses by 0.4–1.0% and $\alpha_e = 1.5$ raises them by 0.5–1.6% over $5 \le M_G \le 13$, with the largest change at $M_G \approx 8$ (Fig. 2b). At $M_G = 4$ both changes are below 0.3%. The metallicity slices [M/H] = −0.6 and +0.3 change by the same amounts to within 0.6%, so the metallicity dependence of the MLR is not affected. At every M_G the change is smaller than the half-width of the 68% credible interval, which is 1–2% at $5 \le M_G \le 10$ and rises to 5% at $M_G = 13$. The outlier fraction and the convergence of the sampler are the same for the three shapes.

5.2. Other modelling choices

6. Discussion

6.1. Limitations

7. Conclusions

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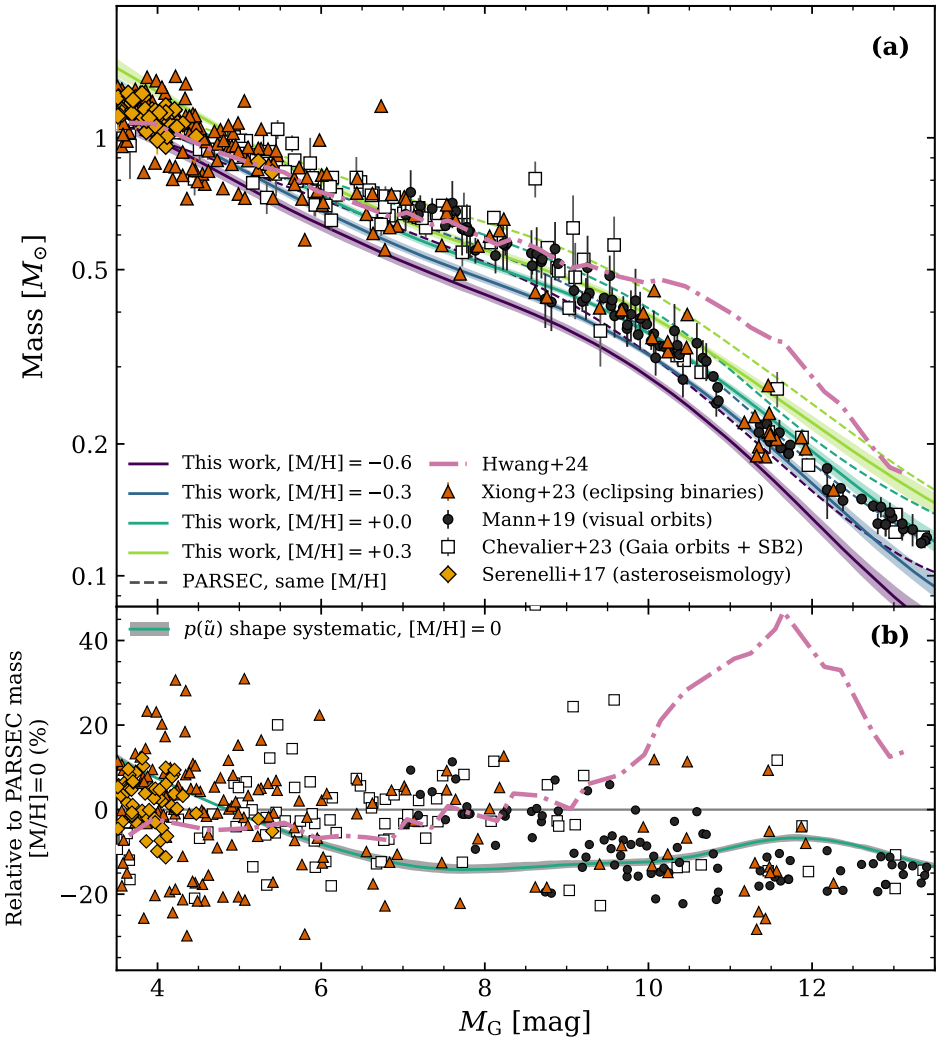


Fig. 2. Dynamical MLR from this work. *Panel (a):* mass as a function of M_G at [M/H] $= -0.6, -0.3, 0, +0.3$ (solid lines, shaded bands: posterior credible intervals), compared with PARSEC at the same metallicities (dashed lines), Hwang et al. (2024) (dash-dotted), and literature masses: eclipsing binaries (Xiong et al. 2023), visual orbits (Mann et al. 2019), Gaia orbits plus SB2 (Chevalier et al. 2023), and asteroseismology (Serenelli et al. 2017). *Panel (b):* mass relative to PARSEC at [M/H] $= 0$. The solid line is the posterior median of this work, and the grey band is the systematic uncertainty from the shape of $p(\tilde{u})$, spanned by the MLRs obtained with eccentricity indices $\alpha_e = 1.0$ and 1.5 around the adopted $\alpha_e = 1.23$ (Sect. 5.1).

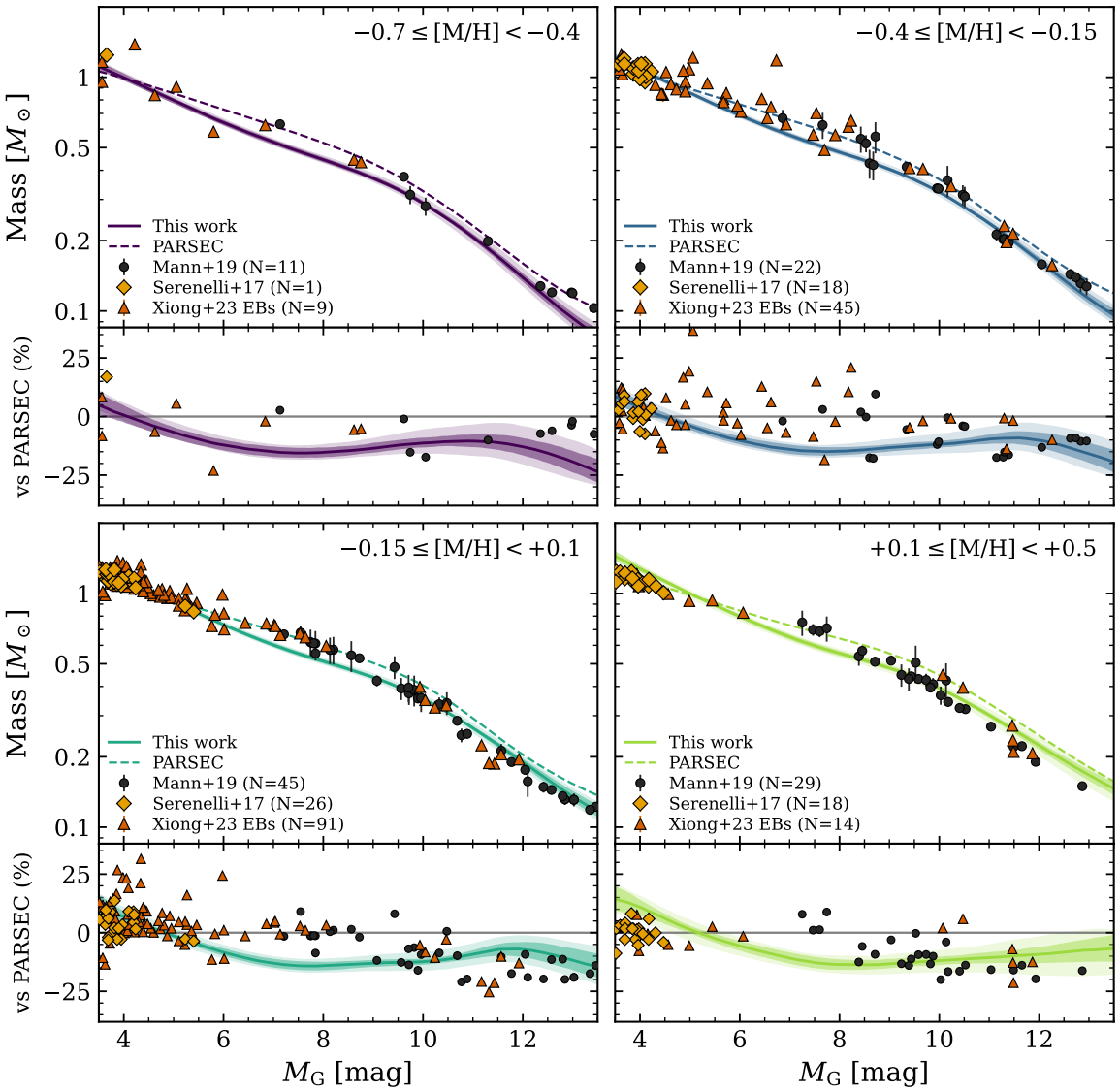


Fig. 3. MLR in four metallicity bins, with the literature calibrators of Fig. 2 selected by [M/H] (sample sizes in the legends). In each bin the upper panel shows the mass and the lower panel the difference from PARSEC at the same metallicity; the solid line is the posterior MLR of this work with credible bands, and the dashed line is PARSEC.

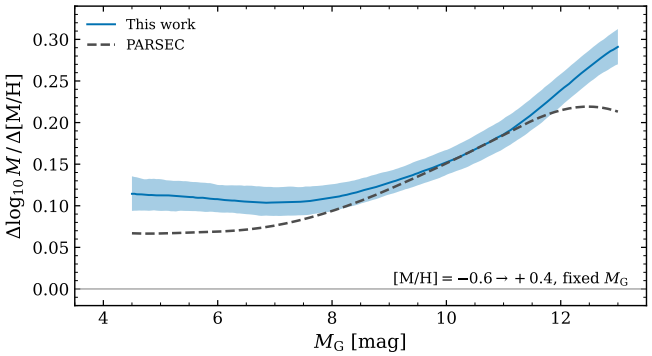


Fig. 4. Metallicity sensitivity of the MLR, $\Delta \log_{10} M/\Delta$[M/H] between [M/H] = −0.6 and +0.4 at fixed M_G. The blue line and band show this work (posterior median and credible interval) and the dashed line shows PARSEC.

Appendix A: Full model specification

Monotone parametrisation. The coefficient grid Θ is generated from unconstrained variables, so that Eq. (5) holds for every draw. With $\mathrm{softplus}(t) = \ln(1 + e^t)$, $\mathrm{sigmoid}(t) = (1 + e^{-t})^{-1}$ and positive scales λ_x, λ_Z, the first row and column are

$$\Theta_{11} = c_0, \tag{A.1}$$
$$\Theta_{1h} = \Theta_{1,h-1} + \lambda_Z \,\mathrm{softplus}(a_h), \quad h = 2, \ldots, 4, \tag{A.2}$$
$$\Theta_{i1} = \Theta_{i-1,1} - \lambda_x \,\mathrm{softplus}(b_i), \quad i = 2, \ldots, 8, \tag{A.3}$$

and the interior points, visited in increasing $i + h$, lie between their two predecessors,

$$\Theta_{ih} = L_{ih} + (U_{ih} - L_{ih})\,\mathrm{sigmoid}(r_{ih}), \tag{A.4}$$
$$L_{ih} = \Theta_{i,h-1}, \quad U_{ih} = \Theta_{i-1,h}. \tag{A.5}$$

If the previously constructed points obey both orders, then $L_{ih} \le \Theta_{i-1,h-1} \le U_{ih}$, so the interval is non-empty and the new point satisfies both inequalities; induction over $i + h$ gives Eq. (5) for the complete grid. The derivative of a B-spline surface is a sum of non-negative basis functions weighted by differences of adjacent coefficients, so the coefficient orders fix the signs of $\partial g/\partial x$ and $\partial g/\partial Z$ everywhere in the domain.

Priors. The parameters and priors of the three stages are listed in Table A.1. The coordinates $(c_0, a_h, b_i, r_{ih}, \log\lambda_x, \log\lambda_Z, f_{\mathrm{outlier}})$ give 35 free parameters. Here $\Pi_{11} = 0.00703$, and $\lambda_{x,\mathrm{ref}} = 0.18624$ and $\lambda_{Z,\mathrm{ref}} = 0.13078$ dex are the mean coefficient increments of the PARSEC surface.

Appendix B: Mock-data validation

Appendix C: Derivation of $p(\widetilde{u})$

The statistic $\widetilde{u}$ depends only on the eccentricity, the orbital phase and the orientation of a pair. The semi-major axis and the distance enter only through the selection of the sample, which acts on the observed projected separation s_{proj} and the distance d. The distribution needed by the likelihood is therefore

$$p(\widetilde{u} \mid \mathrm{sample}) = \int p(\widetilde{u} \mid s_{\mathrm{proj}}, d)\, p_{\mathrm{sample}}(s_{\mathrm{proj}}, d)\, \mathrm{d}s_{\mathrm{proj}}\, \mathrm{d}d. \tag{C.1}$$

We simulate 4×10^6 Keplerian orbits with semi-major axes log-uniform between 10 and 10^5 au, distances log-uniform between 5 and 1000 pc, mean anomalies uniform in $[0, 2\pi)$, isotropic orientations, and eccentricities drawn from $p(e) = (1 + \alpha_e)e^{\alpha_e}$. Kepler's equation is solved for the true anomaly, and the projected separation and projected relative velocity give $\widetilde{u}$ for each orbit. Each simulated pair inside the range of the sample in s_{proj} and d receives the weight $w = H_{\mathrm{sample}}/H_{\mathrm{sim}}$, the ratio of the normalised two-dimensional histograms of the sample and of the simulation in 24 bins of $\log s_{\mathrm{proj}}$ and 18 bins of $\log d$. Because the angular separation is $\theta = s_{\mathrm{proj}}/d$, the weights also reproduce the angular-resolution limit of the catalogue, and an additional cut $\theta > 1.5''$ or $3''$ leaves the fitted constants unchanged. Eq. (8) is fitted by least squares to the weighted histogram of $\widetilde{u}$ in 100 bins on $[0, 40]$. All values of α_e use the same random numbers for the phases, orientations and eccentricity quantiles, so that the differences between the shapes contain no simulation noise.

Hwang et al. (2022) describe the power-law index as a function of $w = \log_{10}(s_{\mathrm{proj}}/\mathrm{au})$ by $\alpha(w) = 1.25\tanh[(w - 1.87)/0.88] + 0.12$. Evaluated at the projected separations of our pairs, it ranges from 1.04 to 1.34 (5th to 95th percentile), because only 1% of the pairs have $s_{\mathrm{proj}} < 300$ au. Drawing one eccentricity for each pair from its $p(e \mid \alpha(w))$ and fitting a single power law to the pooled values by maximum likelihood gives $\alpha_e = 1.23 \pm 0.02$, which we adopt. A simulation with the separation-dependent index applied to each orbit gives a shape with the same median of $\widetilde{u}$ to within 0.02. The fitted constants are listed in Table C.1.

Appendix D: Additional figures and tables

Table A.1. Parameters and priors. Normal distributions are given as location and standard deviation.

Stage	Quantity	Prior	Role
1a	(b_G, s_z, ν)	maximum likelihood	relative calibration; fixed in 1b
1b	δ_z	$\mathcal{N}(0, 0.3)$ dex	J-CaPS–PARSEC zero point
1b	s_C	Half-Normal(0.1) mag	excess colour scatter
1b	$\boldsymbol{\omega}$	Dirichlet$(1, \ldots, 1)$	population weights
2	c_0	$\mathcal{N}(\Pi_{11}, 0.10)$ dex	first coefficient
2	a_h, b_i, r_{ih}	$\mathcal{N}(0, 1)$	monotone coordinates
2	$\log \lambda_x$	$\mathcal{N}(\ln \lambda_{x,\mathrm{ref}}, 0.7)$	increment scale in x
2	$\log \lambda_Z$	$\mathcal{N}(\ln \lambda_{Z,\mathrm{ref}}, 0.7)$	increment scale in Z
2	f_{outlier}	Beta$(3, 12)$	outlier fraction before selection

Table C.1. Constants of Eq. (8) for three eccentricity indices.

α_e	B	u_c	C	median $\widetilde{u}$
1.0	2.2943×10^{-3}	36.2851	3.5567	16.36
1.23	2.5050×10^{-3}	37.2461	2.4500	16.08
1.5	2.6522×10^{-3}	37.9042	1.7151	15.82

Notes. The adopted shape is $\alpha_e = 1.23$. Medians in $\mathrm{km\,s^{-1}}\sqrt{\mathrm{AU}}$.